

- 19 Which one of the following statements about the electric potential and electric field at a point is correct?
- A The potential at the point is always zero when the electric field at that point is zero.
 - B The potential at a point is zero when the electric field around the point is constant.
 - C The electric field at a point is zero when the potential around the point is constant.
 - D The electric potential is given by the rate of change of electric field strength with distance.

Ans: C

Electric field points from a region of higher potential to a region of lower potential. There is no electric field if there is no potential difference i.e. potential is constant.

$$E = -\frac{dV}{dr}$$

If $E = 0$, $\frac{dV}{dr} = 0$ i.e. potential does not change with distance and is hence constant.

20 A high electric potential is applied between two electrodes of a hydrogen discharge tube so that